

MSE 311 Thermodynamics of materials-Homework 13

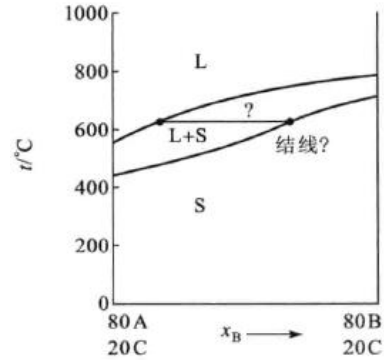
Name:

Student ID:

Please hand in this homework before class on December 15th.

1. Why we cannot arbitrarily mark a horizontal line as a conode on a vertical section as the right figure?

Because the vertical section does not follow the conode of the two-phase region, but intersects with a series of conodes.



2. The probability that a drunk man moves the distance z by n steps is

$$P(n, z) \approx \frac{1}{\sqrt{2\pi n \lambda^2}} \exp\left(-\frac{z^2}{2n \lambda^2}\right)$$

What type of distribution function does this distribution belong to?

3. In a γ -Fe, lattice parameter = 0.37 nm. At 1000°C, $D = 2.5 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$, what the jumping frequency ν_B ?

$$a = \frac{0.37}{\sqrt{2}} \text{ nm}$$

$$\nu_B = \frac{Dz}{a^2} = \frac{2.5 \times 10^{-11} \times 12 \times 2}{0.37^2} = 4.383 \times 10^{-9} \text{ Hz}$$

4. A certain amount of element B was plated on one end of a slender metal rod, and the concentration of element B at 10 μm , 20 μm , 30 μm , 40 μm and 50 μm away from the plating end was 83.8 g/m^3 , 66.4 g/m^3 , 42.0 g/m^3 , 23.6 g/m^3 and 8.74 g/m^3 , respectively, after 24 hours of high temperature annealing. Please estimate simply the diffusion coefficient of B in this metal.

$$C = \frac{2s}{\sqrt{4\pi Dt}} e^{-\frac{y^2}{4Dt}} \Rightarrow \frac{C_2}{C_1} = e^{-\frac{y_1^2 - y_2^2}{4Dt}}$$

$$\frac{y_1^2 - y_2^2}{4Dt} = \ln \frac{C_2}{C_1} \Rightarrow D = \frac{y_1^2 - y_2^2}{4t \ln(C_2 / C_1)}$$

y (μm)	10	20	30	40	50
C (g/m^3)	83.8	66.4	42.0	23.6	8.74
D ($\times 10^{-15}$ m^2/s)		3.73	3.16	3.51	2.62

Average: $3.26 \times 10^{-15} \text{ m}^2/\text{s}$